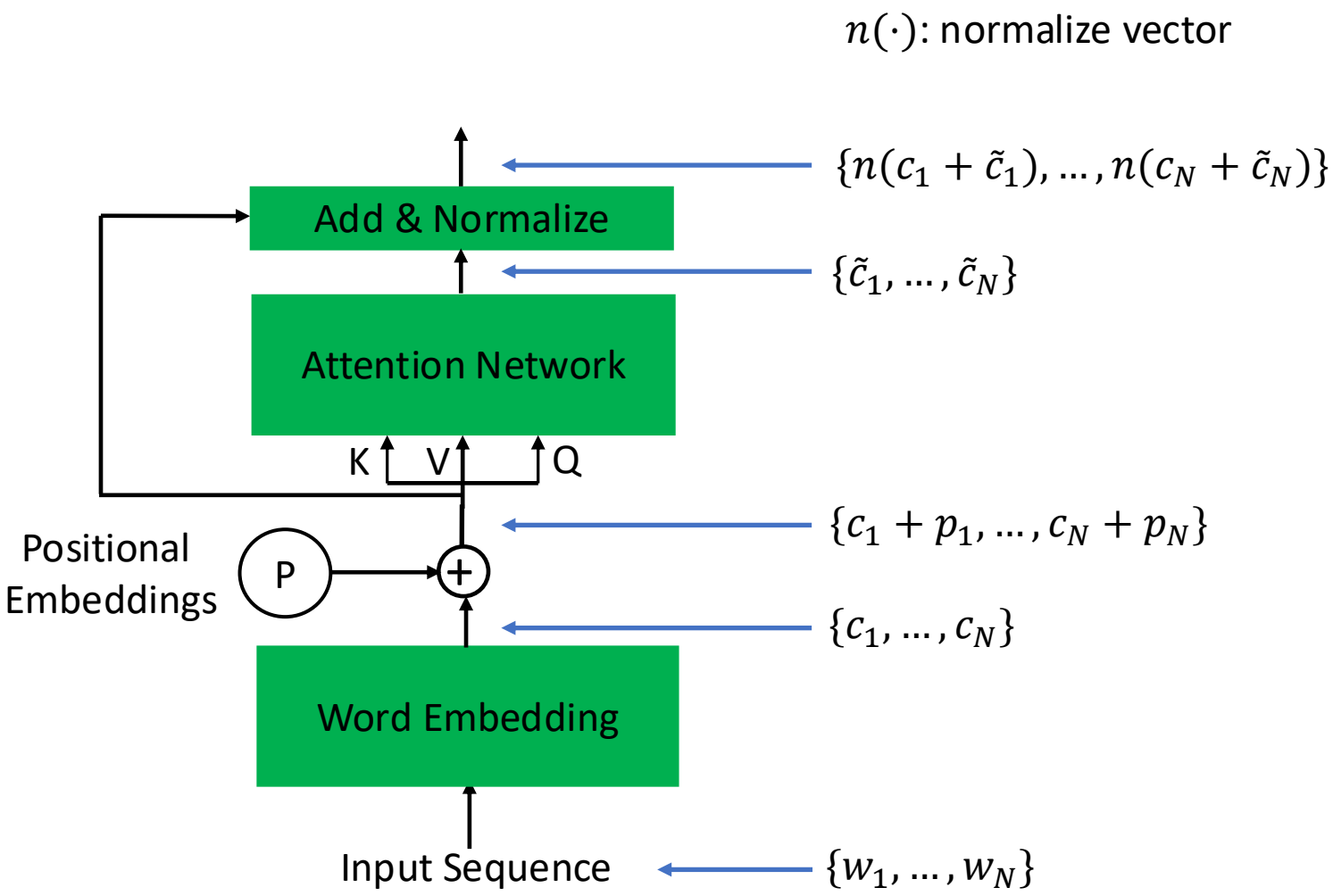
ECE 590
Neural-Network-Based Large Language Models

Tuesdays and Thursdays 1:25-2:40
Fitzpatrick Schiciano A 1464

ECE 590
Neural-Network-Based Large Language Models

Lecture 9
02/05/2026

One Attention Layer



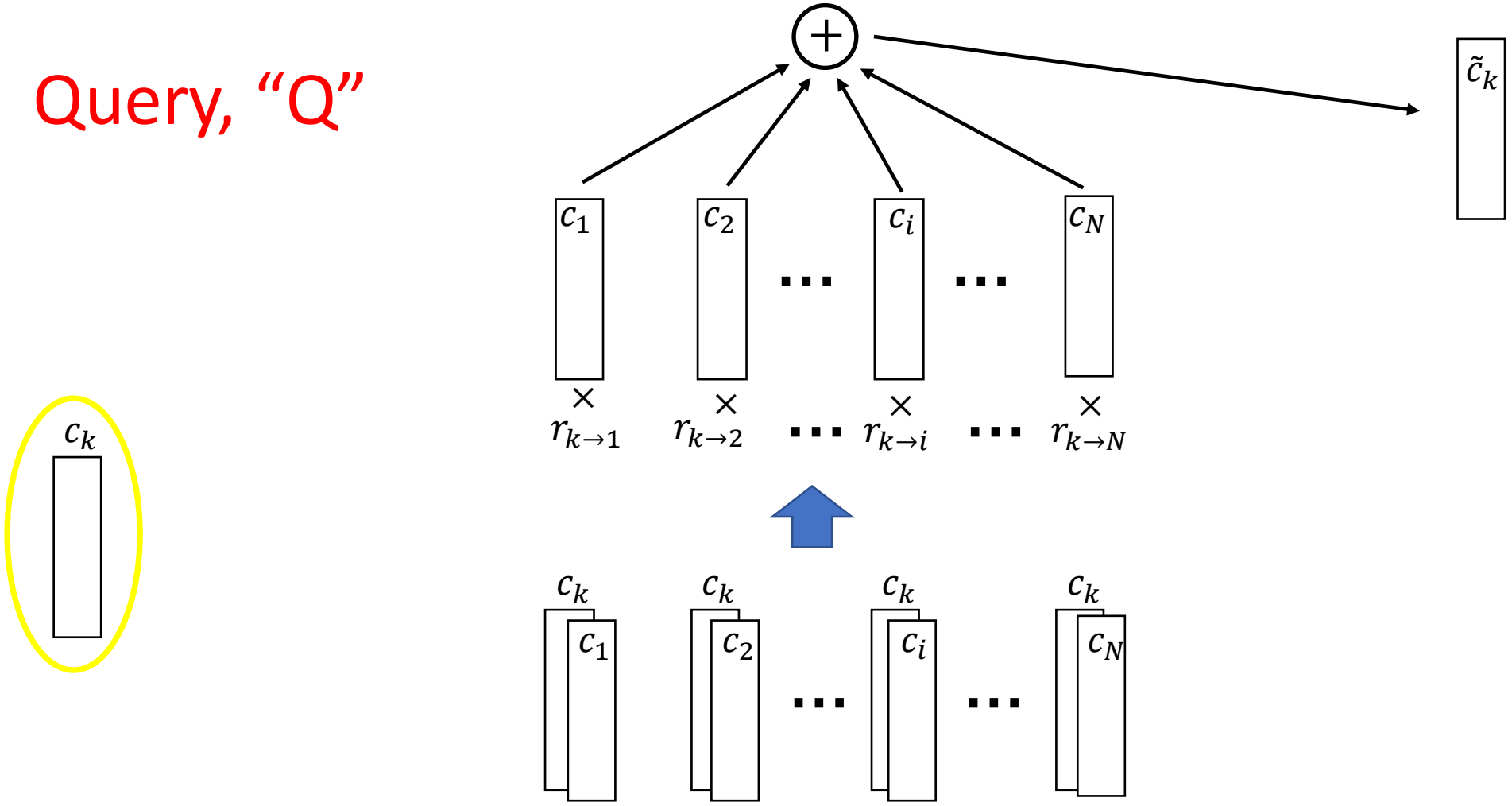
Generalization: “Multi-Headed” Attention

- The encoder and decoder of the Transformer Network are driven by inner products between word vectors

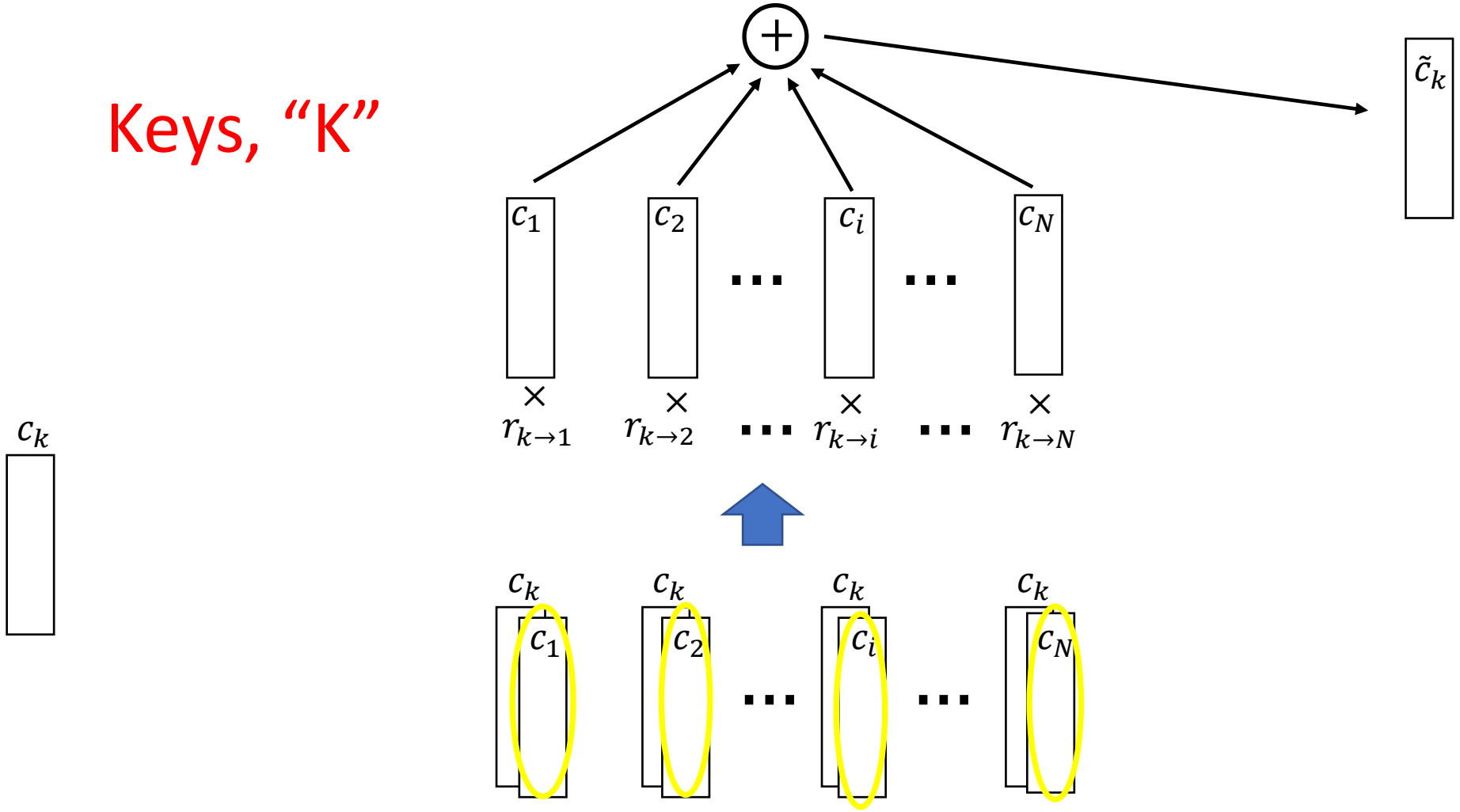
- Associated inner products are performed on word-embeddings vectors

- Generalize the way inner products are performed

Review of Attention Mechanism

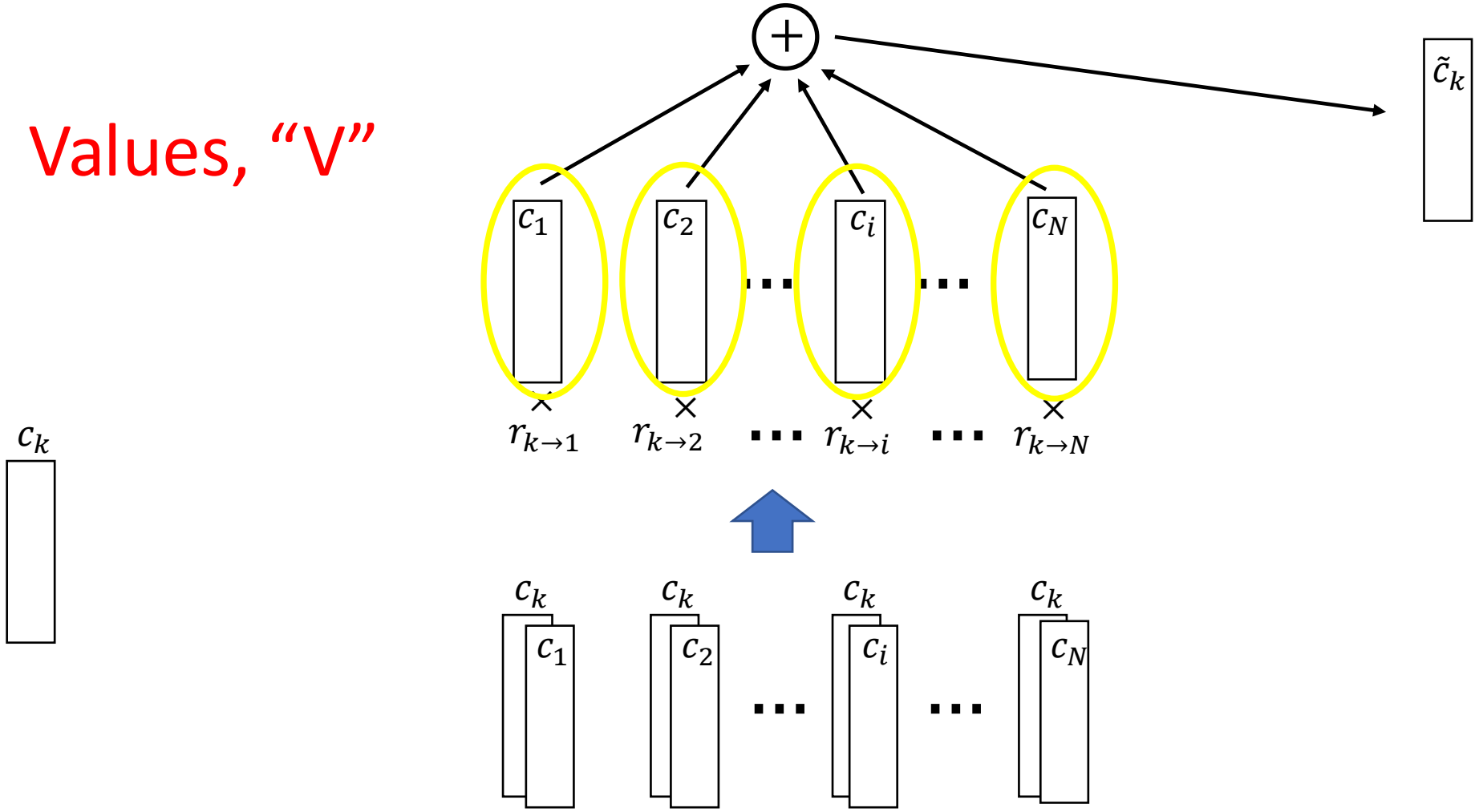


Review of Attention Mechanism



Review of Attention Mechanism

Values, “V”



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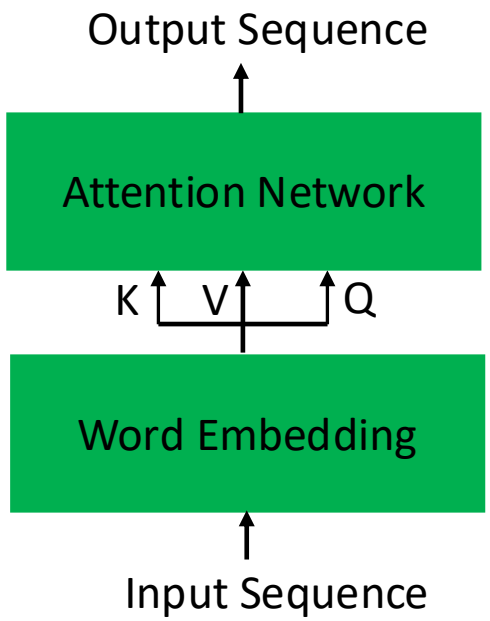

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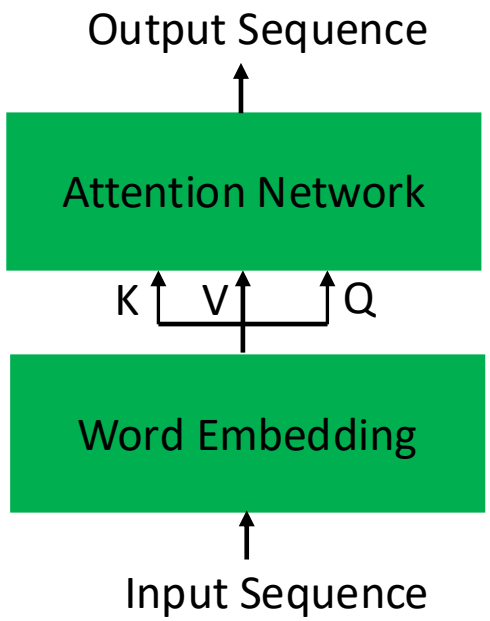
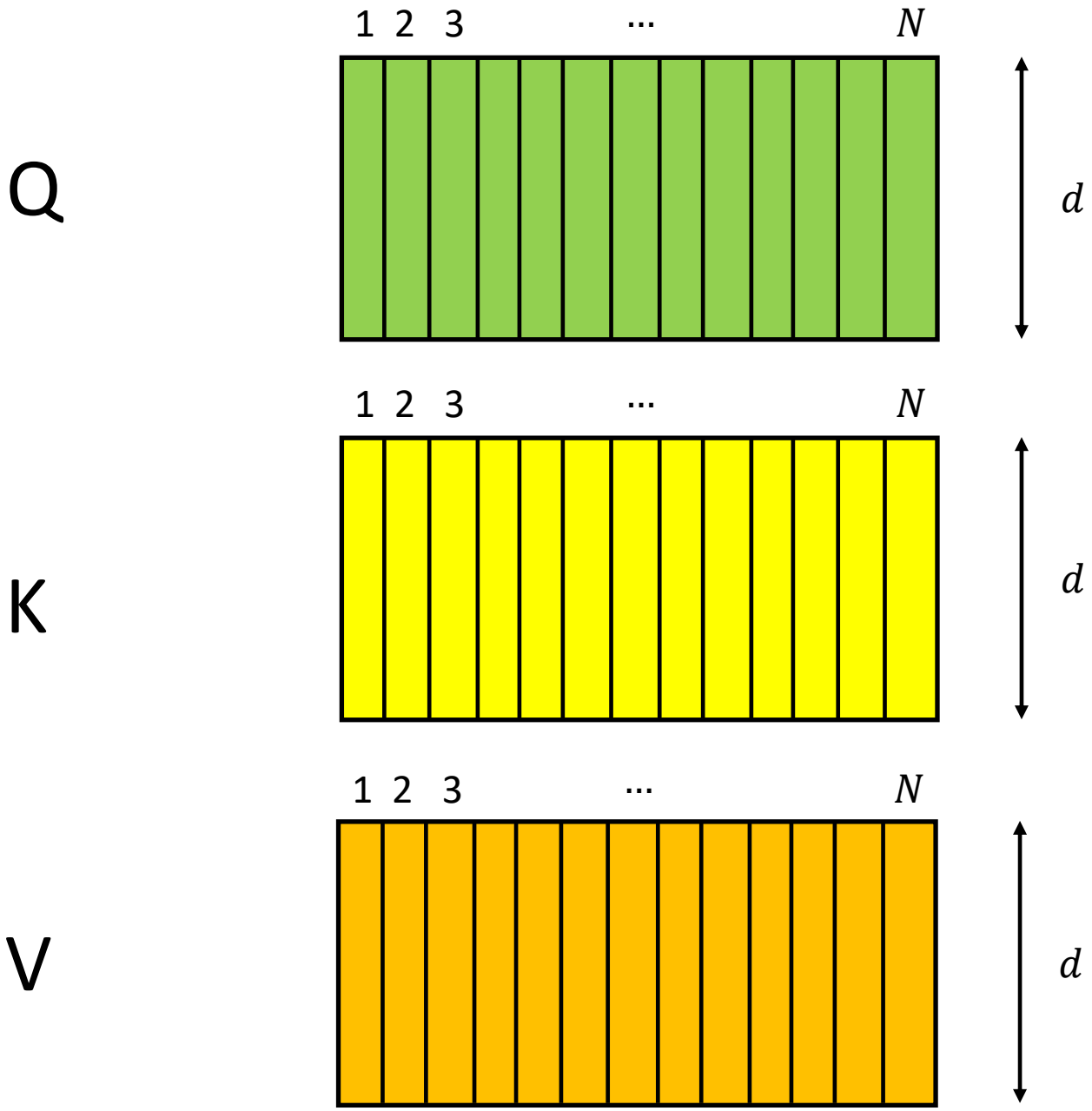


Functional Representation:

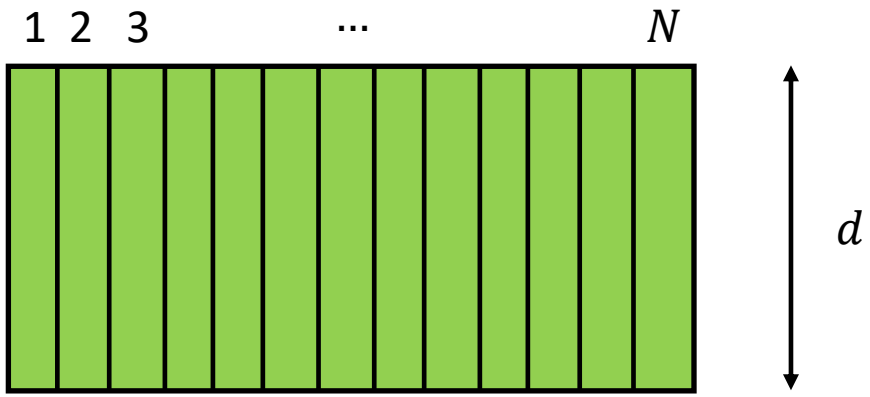
$$\text{output} = \text{Attention}(K, V, Q)$$

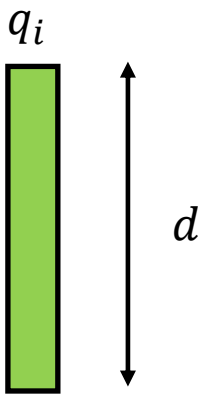
Queries, Keys & Values are Sets of Vectors



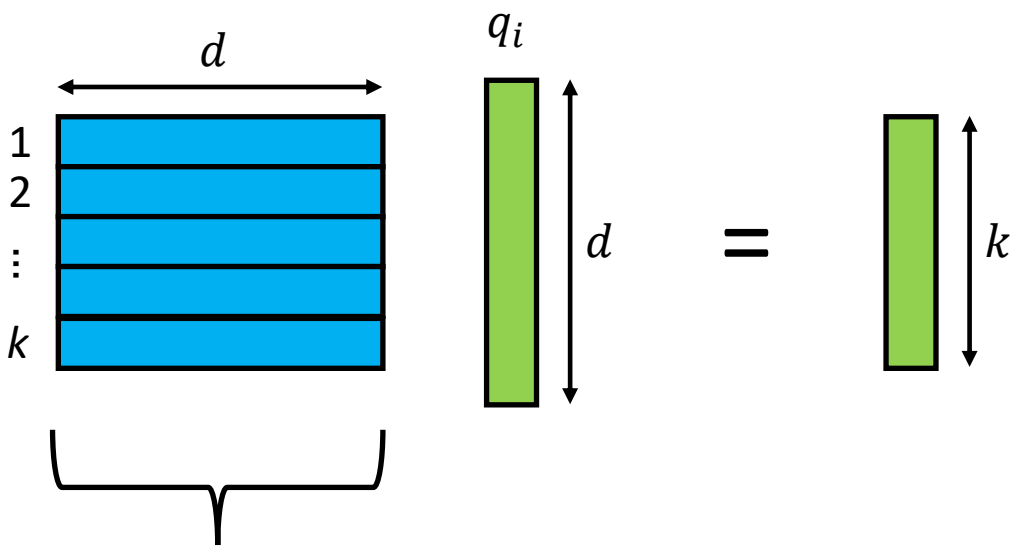
Consider *i*-th Q Vector

Q



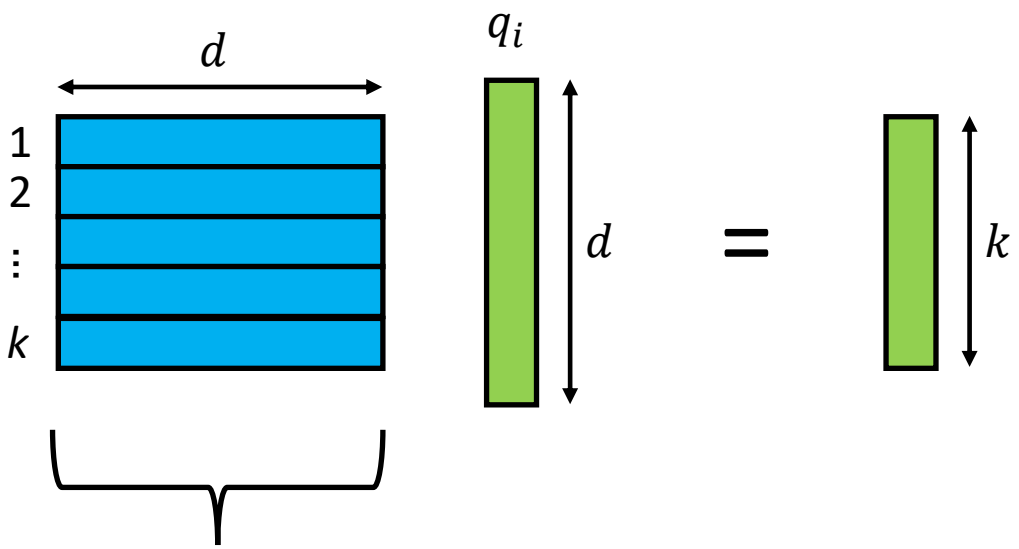


Project Onto k Vectors



Projection Matrix W^Q Associated with Query Vectors

Project Onto *k* Vectors

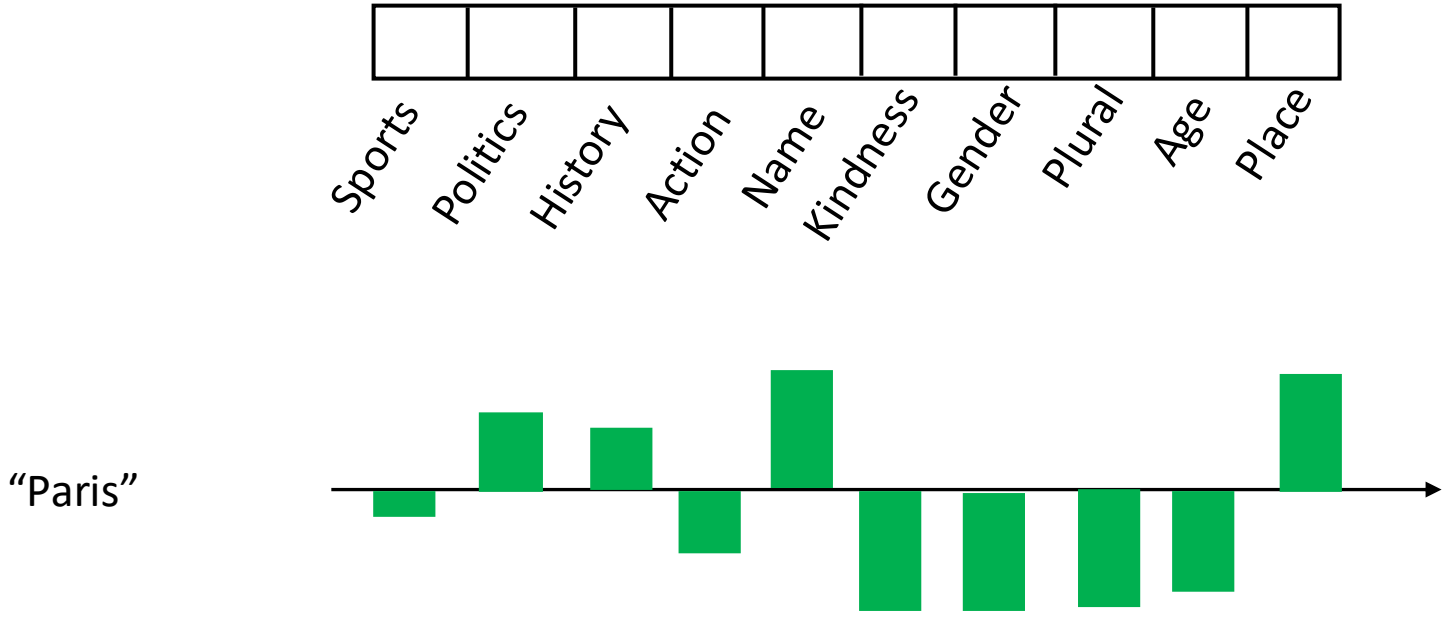


Projection Matrix W^Q Associated with Query Vectors

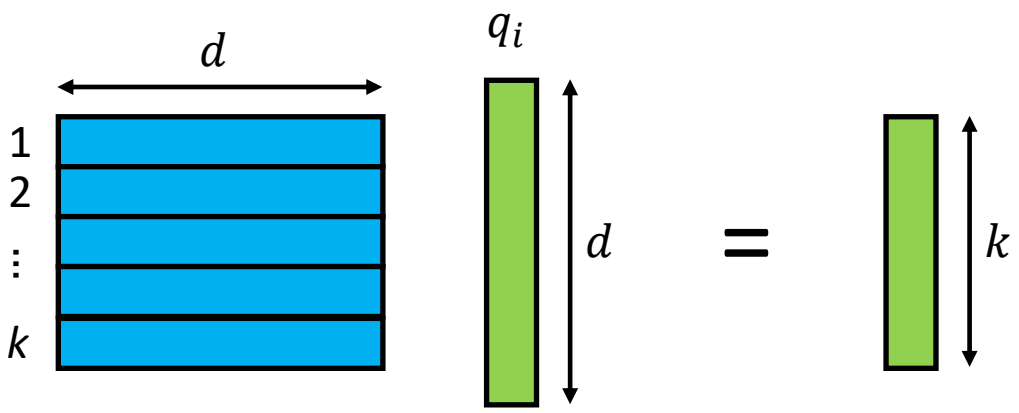
- ❑ Maps query vector into a *k*-dimensional linear subspace defined by rows of matrix W^Q
- ❑ The new, projected vector denoted $W^Q q_i$, and as applied to all queries as $W^Q Q$

Interpretation

- Recall the intuition behind word vectors

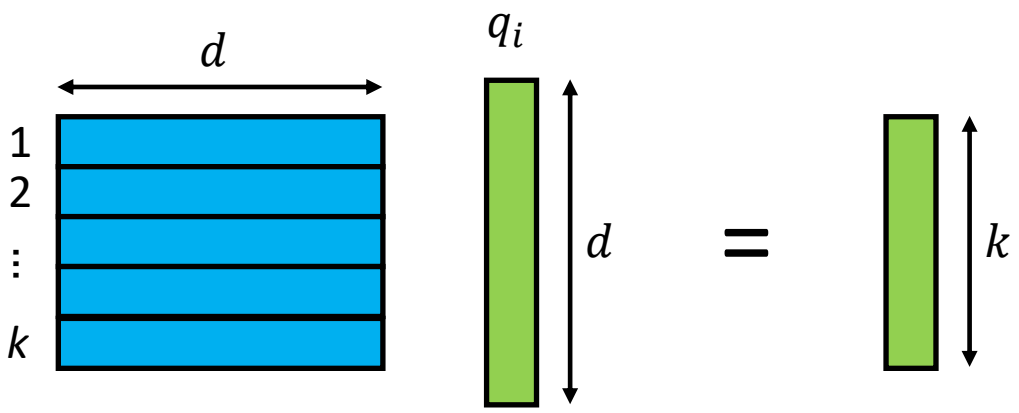


Interpretation



- k components of output correspond to performing k inner products with q_i

Interpretation



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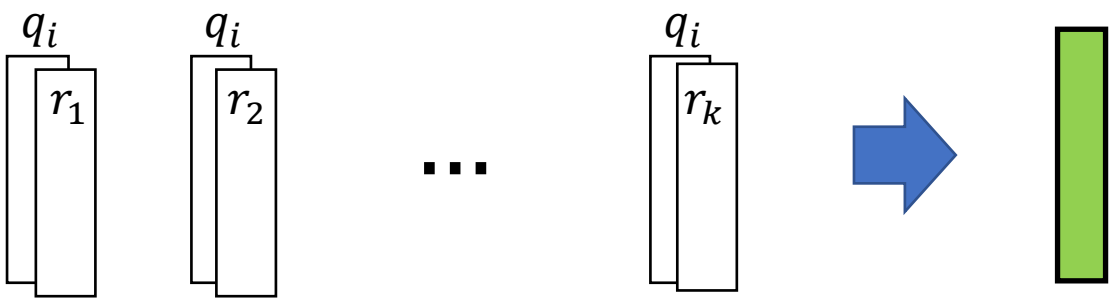
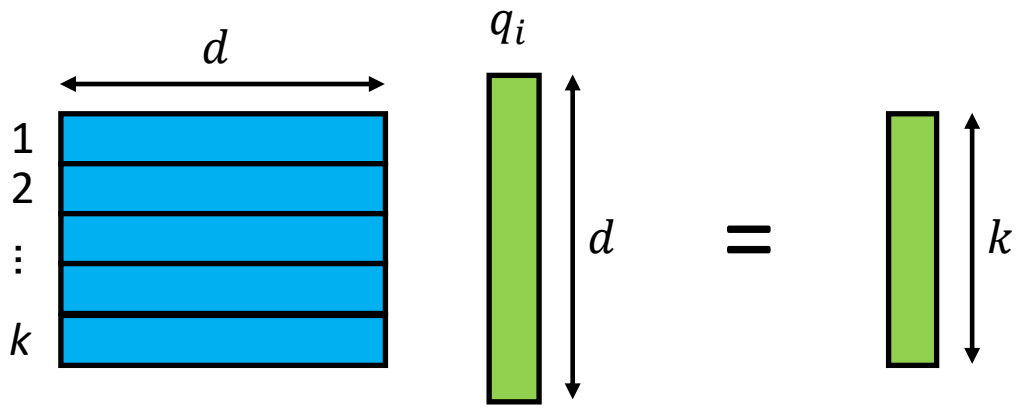
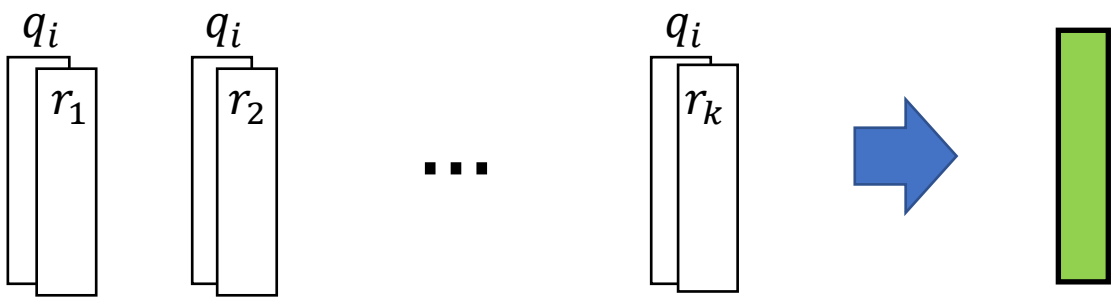


$r_1, r_2, \ldots, r_k$ represent the k rows of W^Q

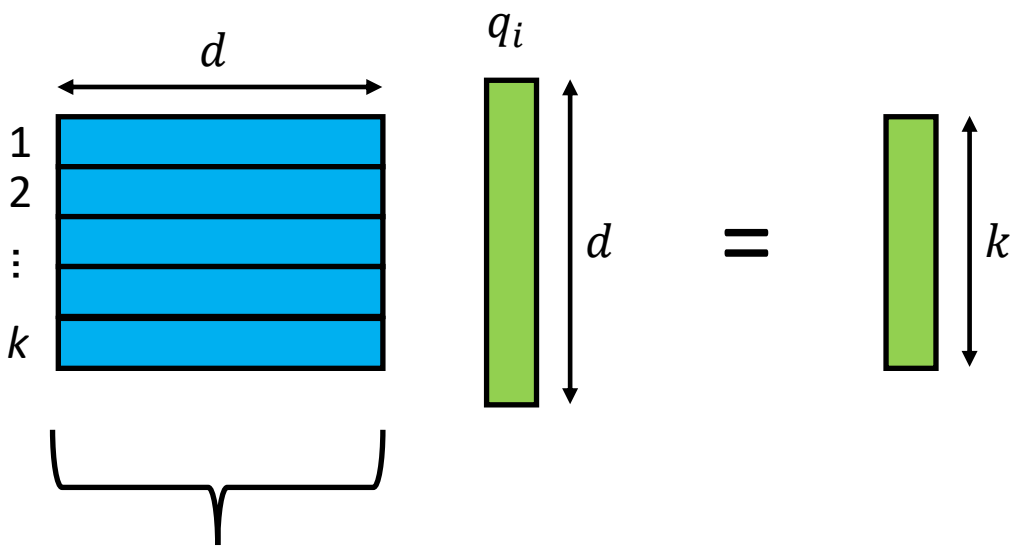
Interpretation





- ❑ Each row r_j corresponds to a particular weighting of word-vector "topics", *i.e.*, "meta topics"
- ❑ Map each word vector into new vector representative of meta-topics

h Projection "Heads"

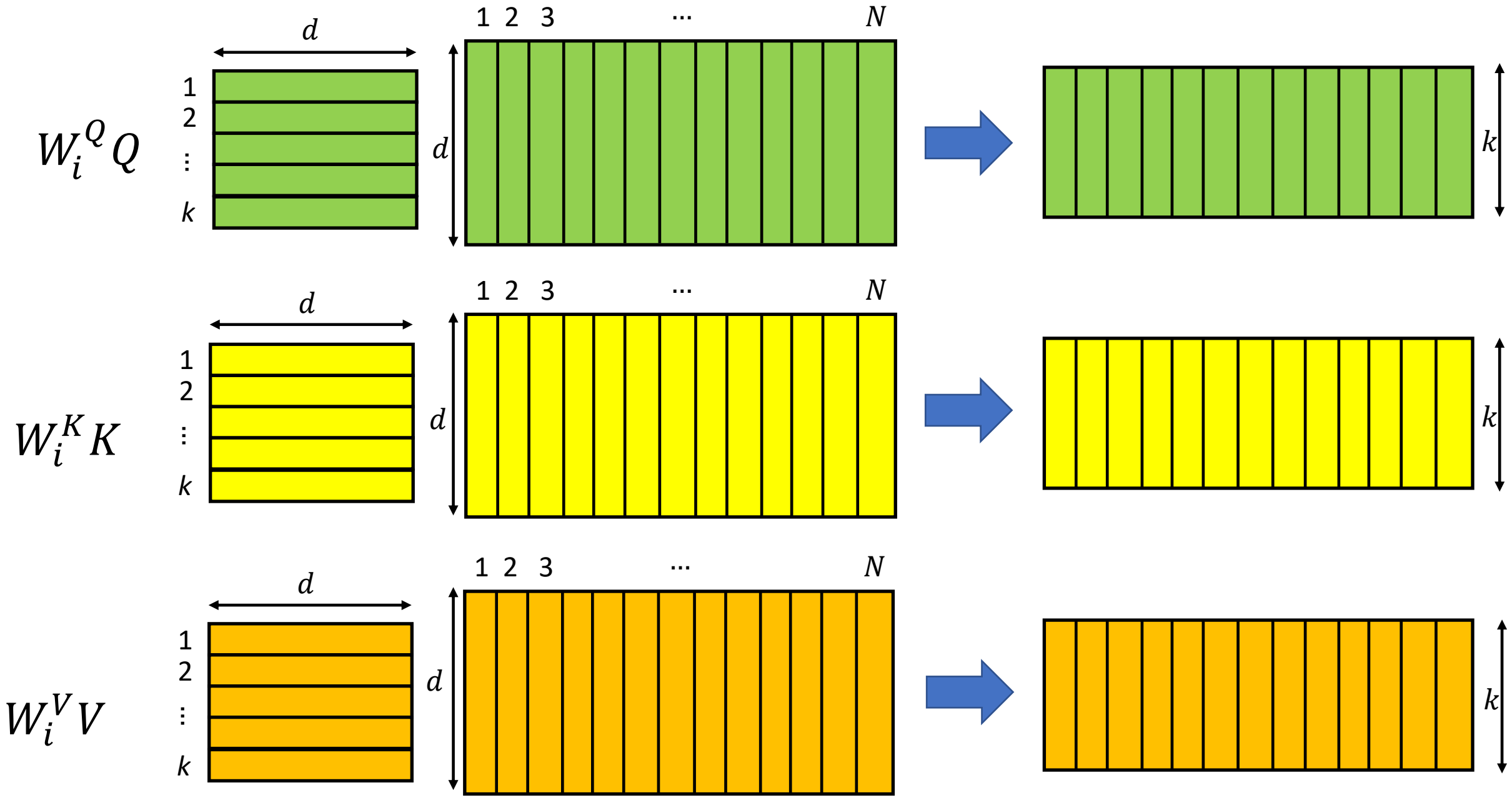


i-th Projection Matrix W_i^Q Associated with Query Vectors

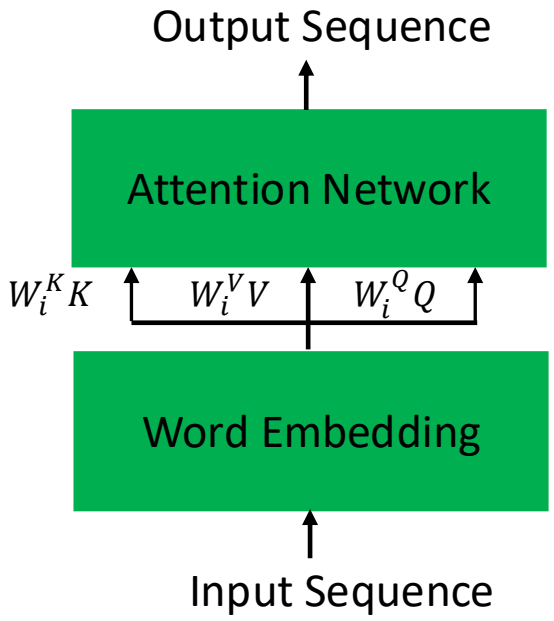
- Constitute *h* matrices of this type: $W_1^Q, W_2^Q, \ldots, W_h^Q$

- These matrices project query vectors into *h* linear subspaces, emphasizing different parts of the query vectors

i-th Projection Matrices for Q, K and V

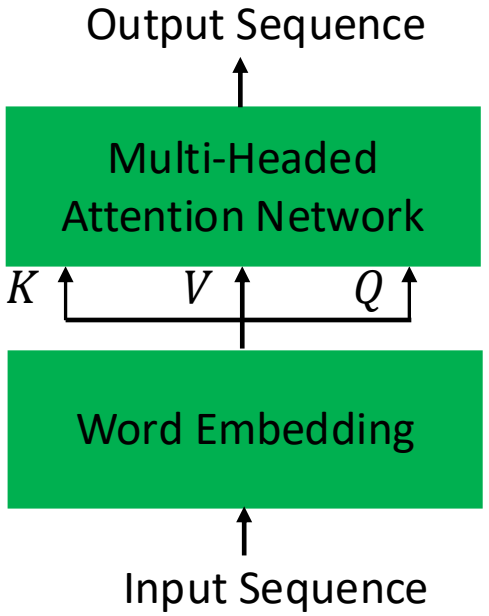


Attention for Head *i*



$$\text{output}_i = \text{Attention}(W_i^K K, W_i^V V, W_i^Q Q)$$

Multi-Headed Attention



$$\text{Multi} - \text{Headed Output} = \text{Concat}(\text{output}_1, \dots, \text{output}_h)W^O$$

$$\text{output}_i = \text{Attention}(W_i^K K, W_i^V V, W_i^Q Q)$$

Multi-Headed Attention: Motivation

$$\text{output}_i = \text{Attention}(W_i^K K, W_i^V V, W_i^Q Q)$$

$$\text{Multi} - \text{Headed Output}_i = \text{Concat}(\text{output}_1, \dots, \text{output}_h) W^O$$

- The h projections associated with the multi-headed attention allow the model to attend based on different aspects (projections) of the word vectors

- Allows attention to focus on different aspects of the language/words

- Introduces many new parameters: the projection matrices

More on Multi-Head Attention & Positional Embeddings

- Recall the word embedding plus positional embedding, for word n in the sequence:

$$w_n = e_n + p_n$$

- For head i, the multi-head attention acts on this sum as (because of linearity):

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

Positional Embeddings

Word Position

Embedding Dimension 1

Embedding Dimension 2

Embedding Dimension 3

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.

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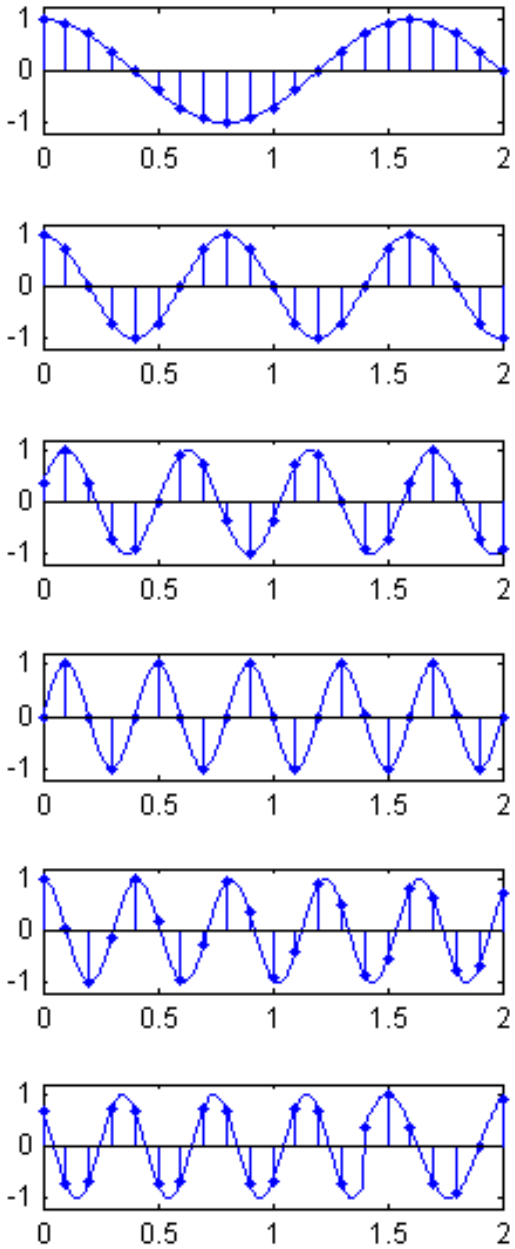

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d})$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d})$$

More on Multi-Head Attention & Positional Embeddings

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- Recall that the positional embeddings have components that oscillate at different frequencies
- The head matrices W_i^Q, W_i^K and W_i^V may be learned to emphasize different components of the positional embeddings, effectively setting a measure of scale of closeness (proximity)
- The head matrices may also emphasize different components of e_n, thereby emphasizing particular topics or writing styles

Example of What Multiple Heads Allow

- Direct particular attention to words <u>nearby next word</u> to be predicted

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n \approx W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q e_n \approx 0_d$ and $W_i^K e_n \approx 0_d$ where embeddings of length d and 0_d is a d-dimensional vector of all zeros

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of p_n that correspond to appropriate scales of positional "closeness" (recall components oscillate with position at frequency that depends on the positional-encoding component)

Example of What Multiple Heads Allow

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n \approx W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q e_n \approx 0_d$ and $W_i^K e_n \approx 0_d$ where embeddings of length d and 0_d is a d-dimensional vector of all zeros

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of p_n that correspond to appropriate scales of positionally "close" (recall components oscillate with position at frequency that depends on the positional-encoding component)

Example of What Multiple Heads Allow

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q p_n$$

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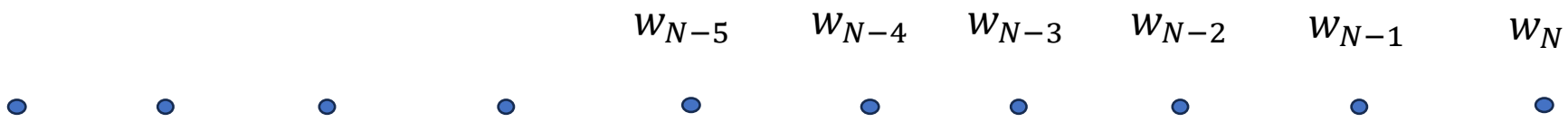
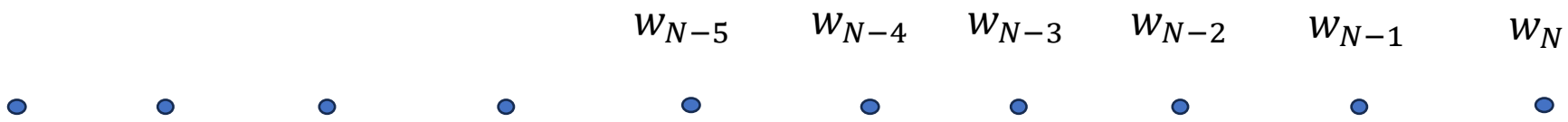
$$W_i^V w_n = W_i^V e_n + W_i^V p_n \approx W_i^V e_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q e_n \approx 0_d$ and $W_i^K e_n \approx 0_d$ where embeddings of length d and 0_d is a d-dimensional vector of all zeros

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of p_n that correspond to appropriate scales of positionally "close" (recall components oscillate with position at frequency that depends on the positional-encoding component)

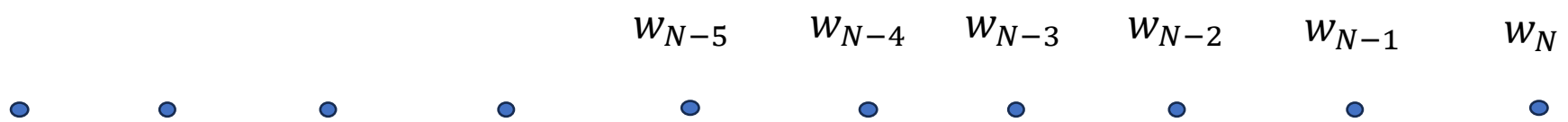
- W_i^V learned such that $W_i^V p_n \approx 0_d$ and $W_i^V e_n \neq 0_d$ and rows of W_i^V emphasize particular aspects of word meaning, of "most relevance" when considering nearby words


w_{N-5}
w_{N-4}
w_{N-3}
w_{N-2}
w_{N-1}
w_N



Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

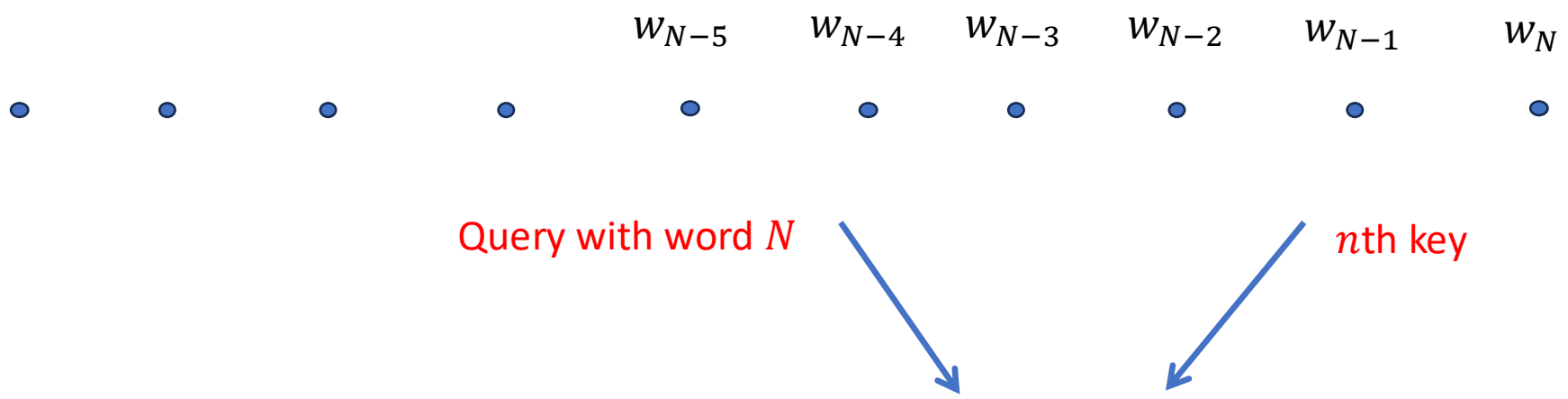


Query with word N

n-th key

Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$



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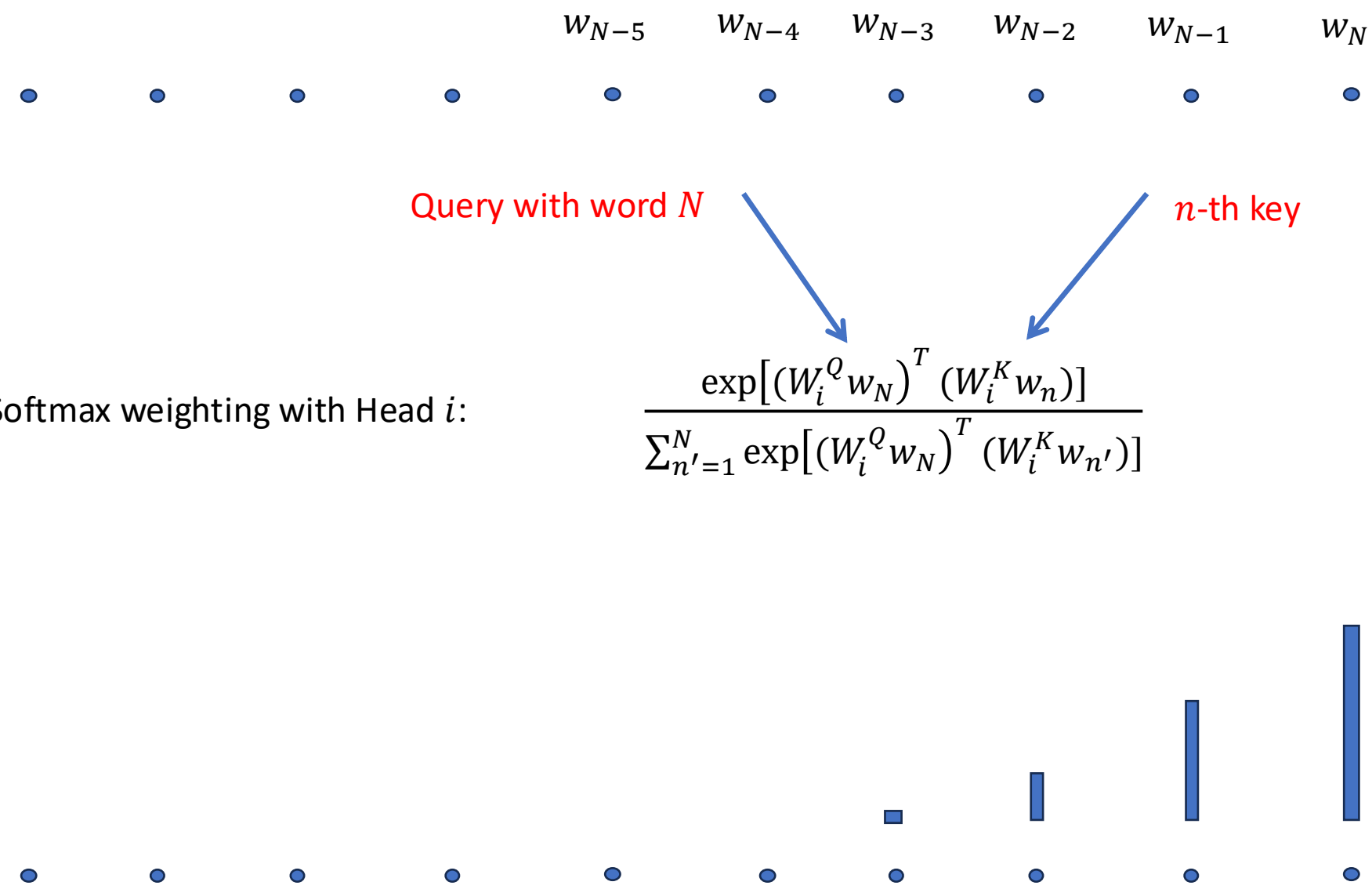
i-th attention head

- W_i^Q and W_i^K designed (learned) such that $W_i^Q e_n \approx 0_d$ and $W_i^K e_n \approx 0_d$ where embeddings of length d and 0_d is a d-dimensional vector of all zeros

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of p_n that correspond to appropriate scales of positionally "close" (recall components oscillate with position at frequency that depends on the positional-encoding component)

- W_i^V learned such that $W_i^V p_n \approx 0_d$ and $W_i^V e_n \neq 0_d$ and rows of W_i^V emphasize particular aspects of word meaning, of "most relevance" when considering nearby words



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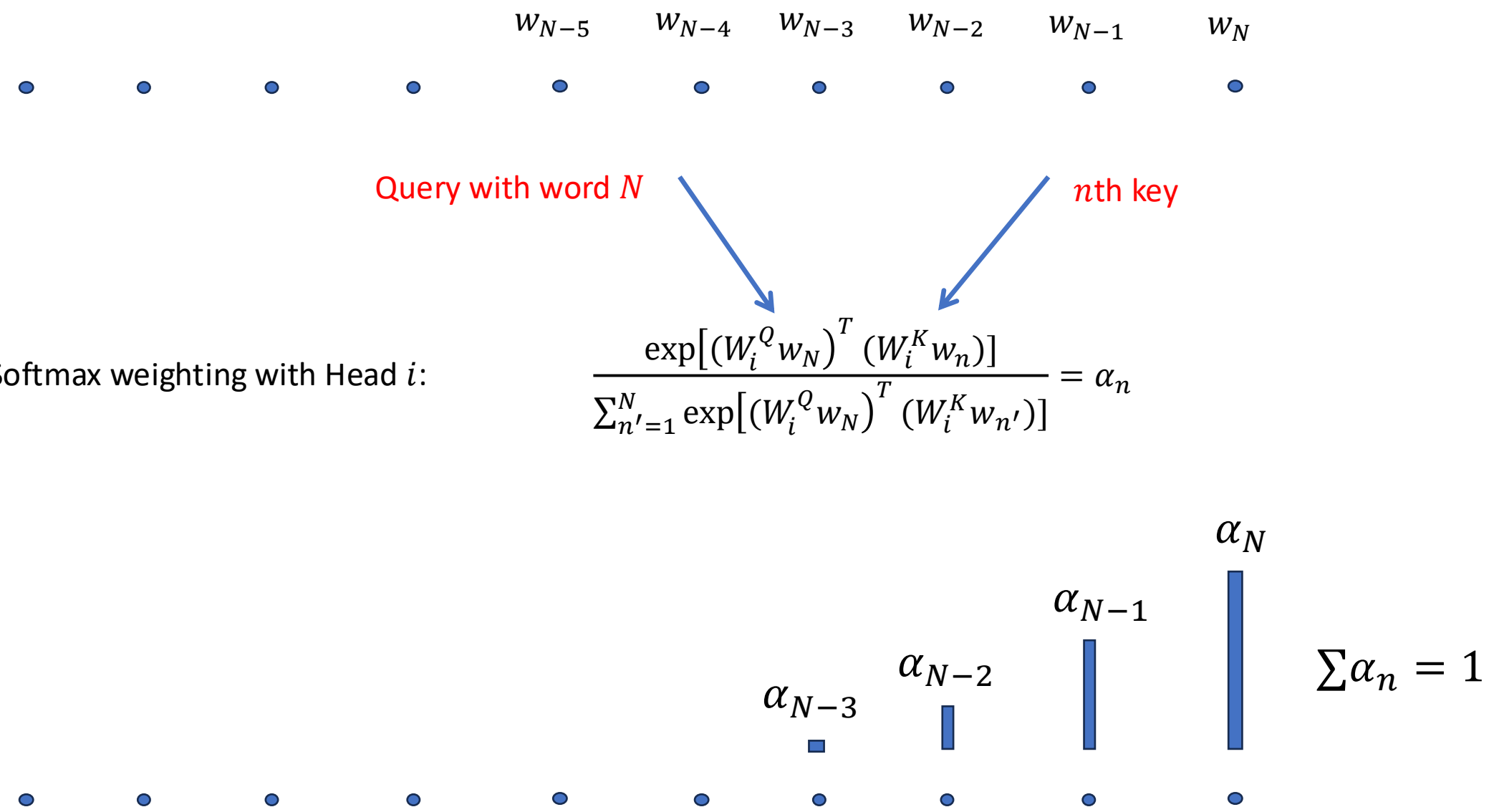
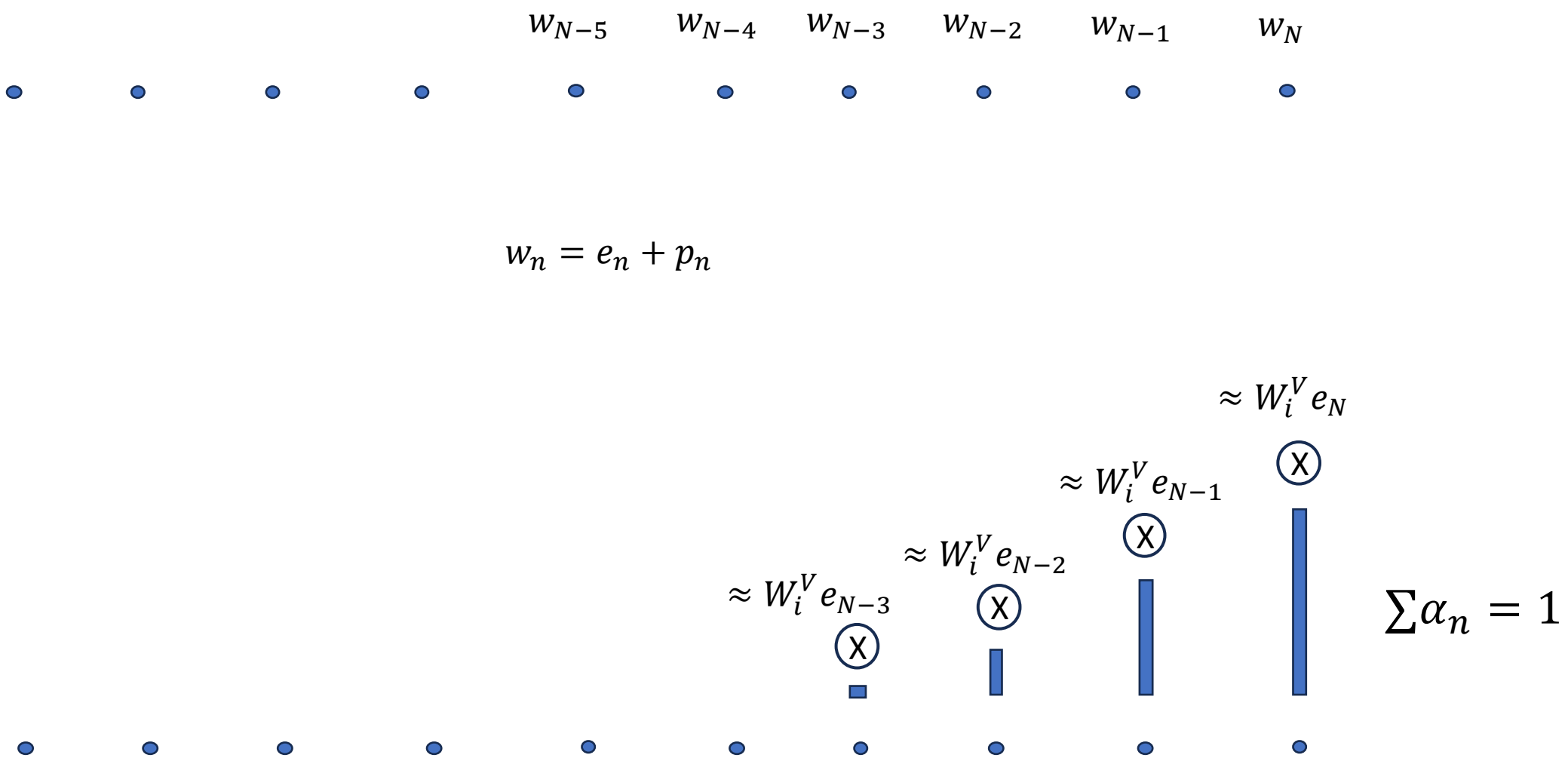
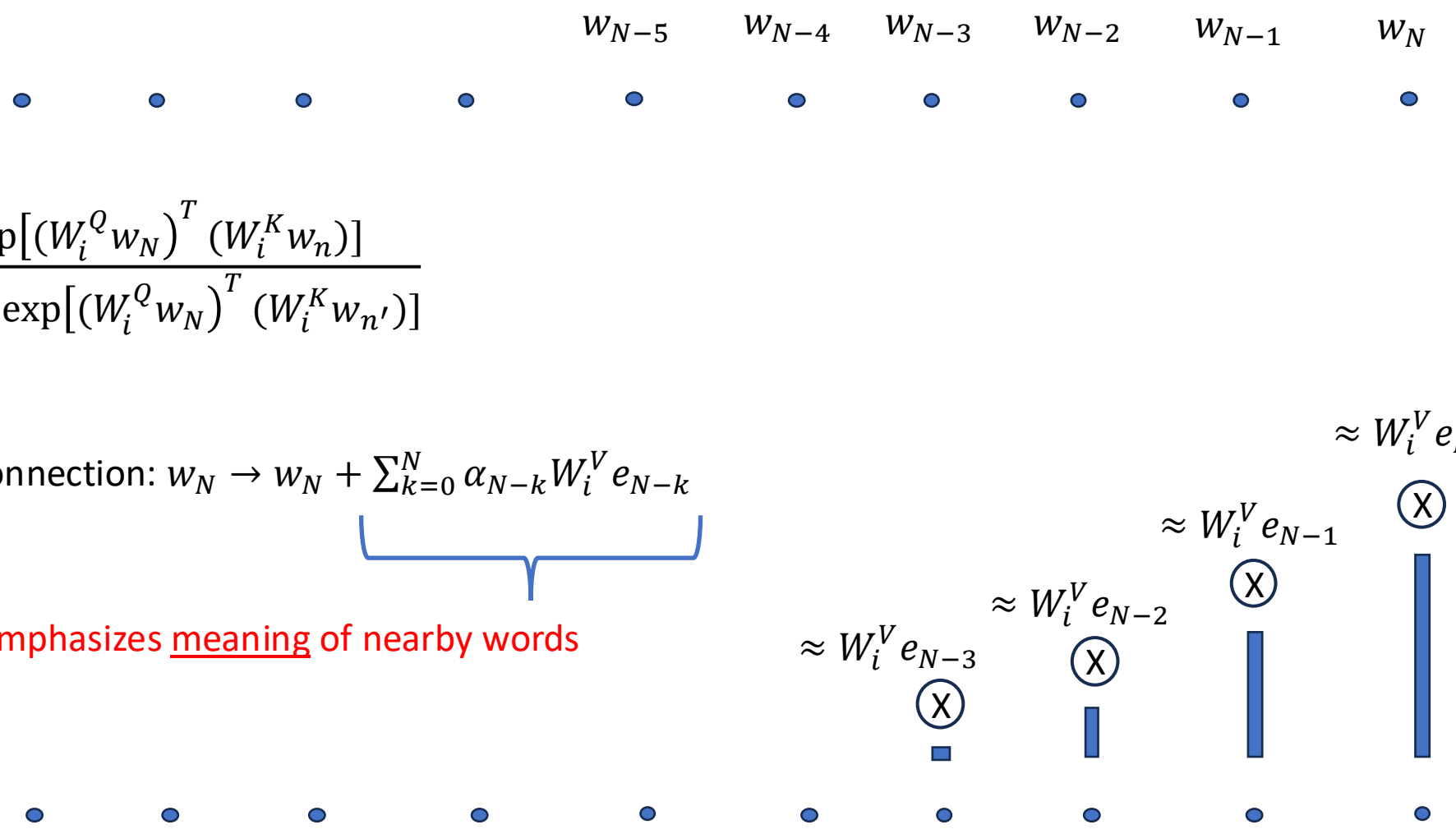


Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]} = \alpha_n$$


w_{N-5}
w_{N-4}
w_{N-3}
w_{N-2}
w_{N-1}
w_N
$w_n = e_n + p_n$
$\approx W_i^V e_N$
$\approx W_i^V e_{N-1}$
$\approx W_i^V e_{N-2}$
$\approx W_i^V e_{N-3}$
$\sum \alpha_n = 1$

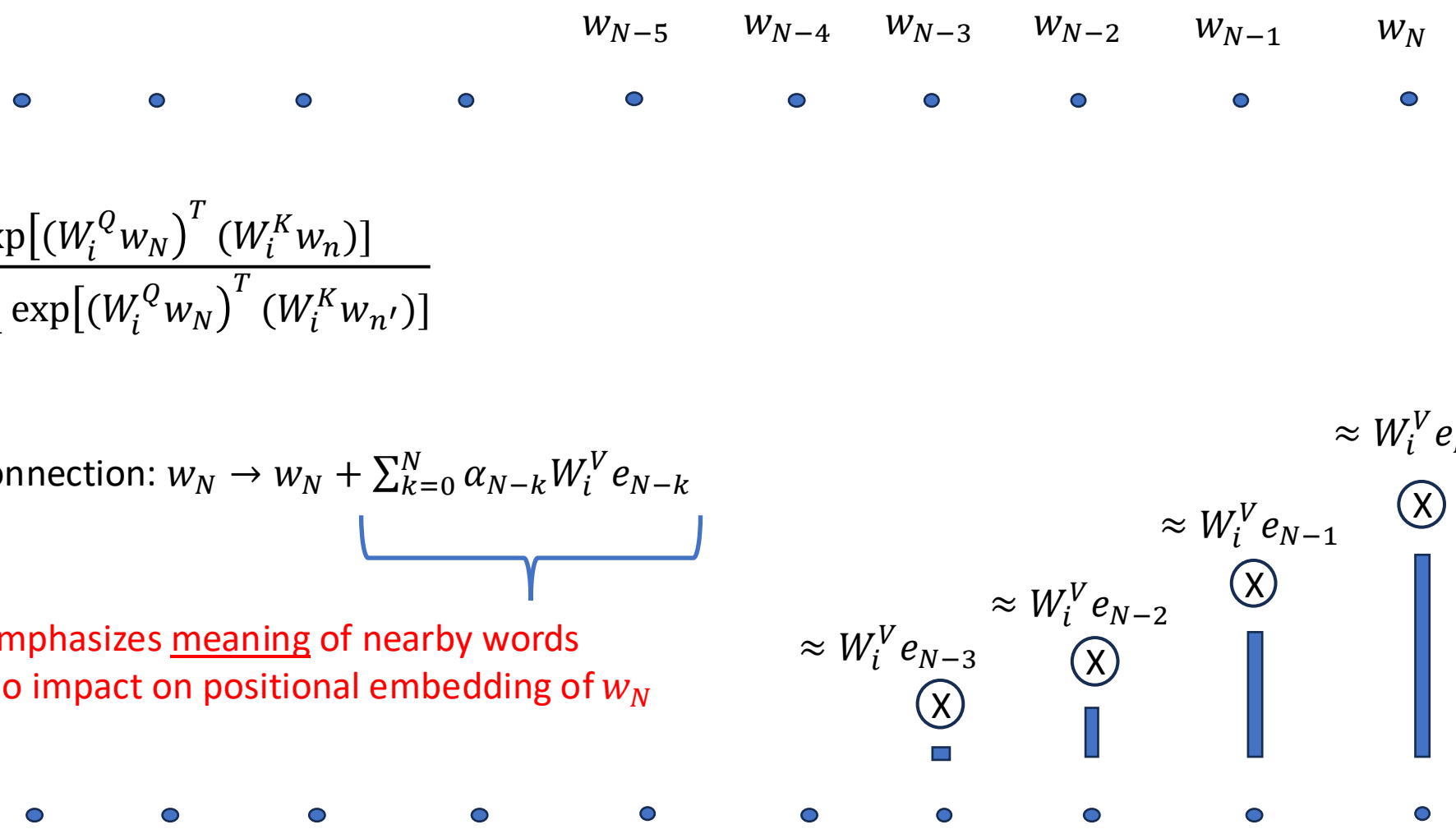


$$\alpha_N = \frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

With skip connection: $w_N \rightarrow w_N + \underbrace{\sum_{k=0}^{N} \alpha_{N-k} W_i^V e_{N-k}}$

Emphasizes <u>meaning</u> of nearby words

$$\sum \alpha_n = 1$$



$$\alpha_N = \frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

With skip connection: $w_N \rightarrow w_N + \underbrace{\sum_{k=0}^{N} \alpha_{N-k} W_i^V e_{N-k}}$

Emphasizes <u>meaning</u> of nearby words

No impact on positional embedding of w_N

$$\sum \alpha_n = 1$$

Example of What Multiple Heads Allow

- Direct particular attention to words of relevant meaning to query (e.g., if query is a verb, emphasize proper nouns/names)

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q e_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n \approx W_i^K e_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q p_n \approx 0_d$ and $W_i^K p_n \approx 0_d$ and therefore position <u>not</u> emphasized for query-key match, but rather meaning relatedness emphasized

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of e_n (word meaning) that are most relevant to relatedness, recalling that components of e_n contain meaning

Example of What Multiple Heads Allow

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q e_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n \approx W_i^K e_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q p_n \approx 0_d$ and $W_i^K p_n \approx 0_d$ and therefore position <u>not</u> emphasized for query-key match, but rather meaning relatedness emphasized

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of e_n (word meaning) that are most relevant to relatedness, recalling that components of e_n contain meaning

Example of What Multiple Heads Allow

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n \approx W_i^Q e_n$$

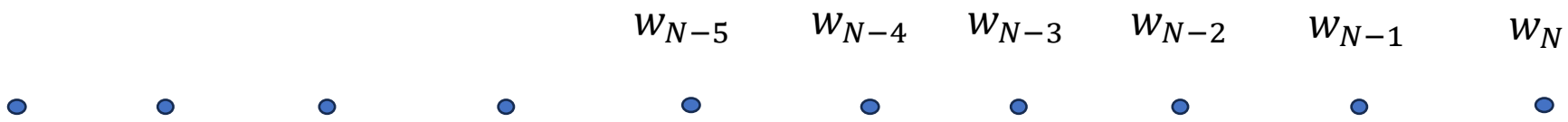
$$W_i^K w_n = W_i^K e_n + W_i^K p_n \approx W_i^K e_n$$

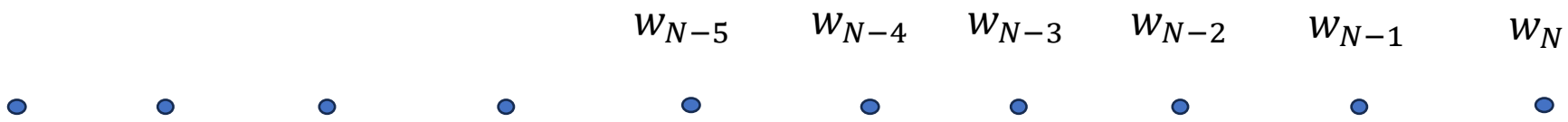
$$W_i^V w_n = W_i^V e_n + W_i^V p_n \approx W_i^V e_n$$

- W_i^Q and W_i^K designed (learned) such that $W_i^Q p_n \approx 0_d$ and $W_i^K p_n \approx 0_d$ and therefore position <u>not</u> emphasized for query-key match, but rather meaning relatedness emphasized

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of e_n (word meaning) that are most relevant to relatedness, recalling that components of e_n contain meaning

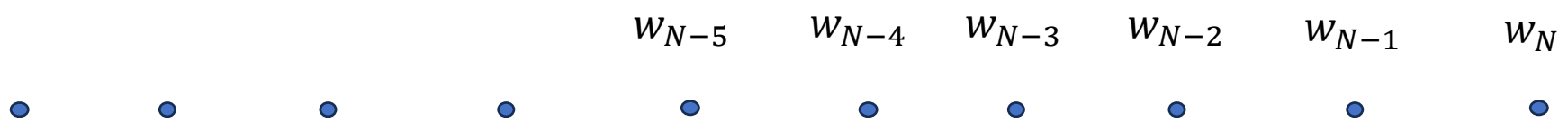
- W_i^V learned such that $W_i^V p_n \approx 0_d$ and $W_i^V e_n \neq 0_d$ and rows of W_i^V emphasize particular aspects of word meaning, of "most relevance", *e.g.*, query may emphasize if word a noun, and key emphasizes if word a verb, or adjectives related to nouns


w_{N-5}
w_{N-4}
w_{N-3}
w_{N-2}
w_{N-1}
w_N



Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

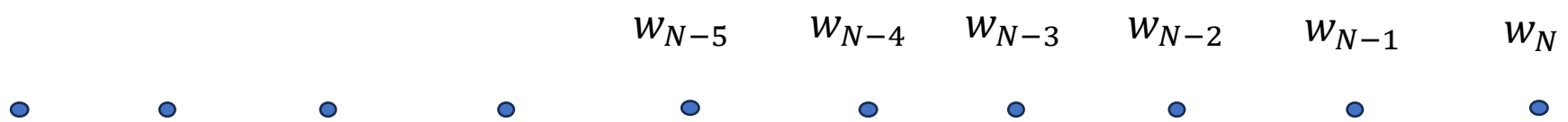


Query with word N

n-th key

Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$



Query with word N

nth key

Softmax weighting with Head i:

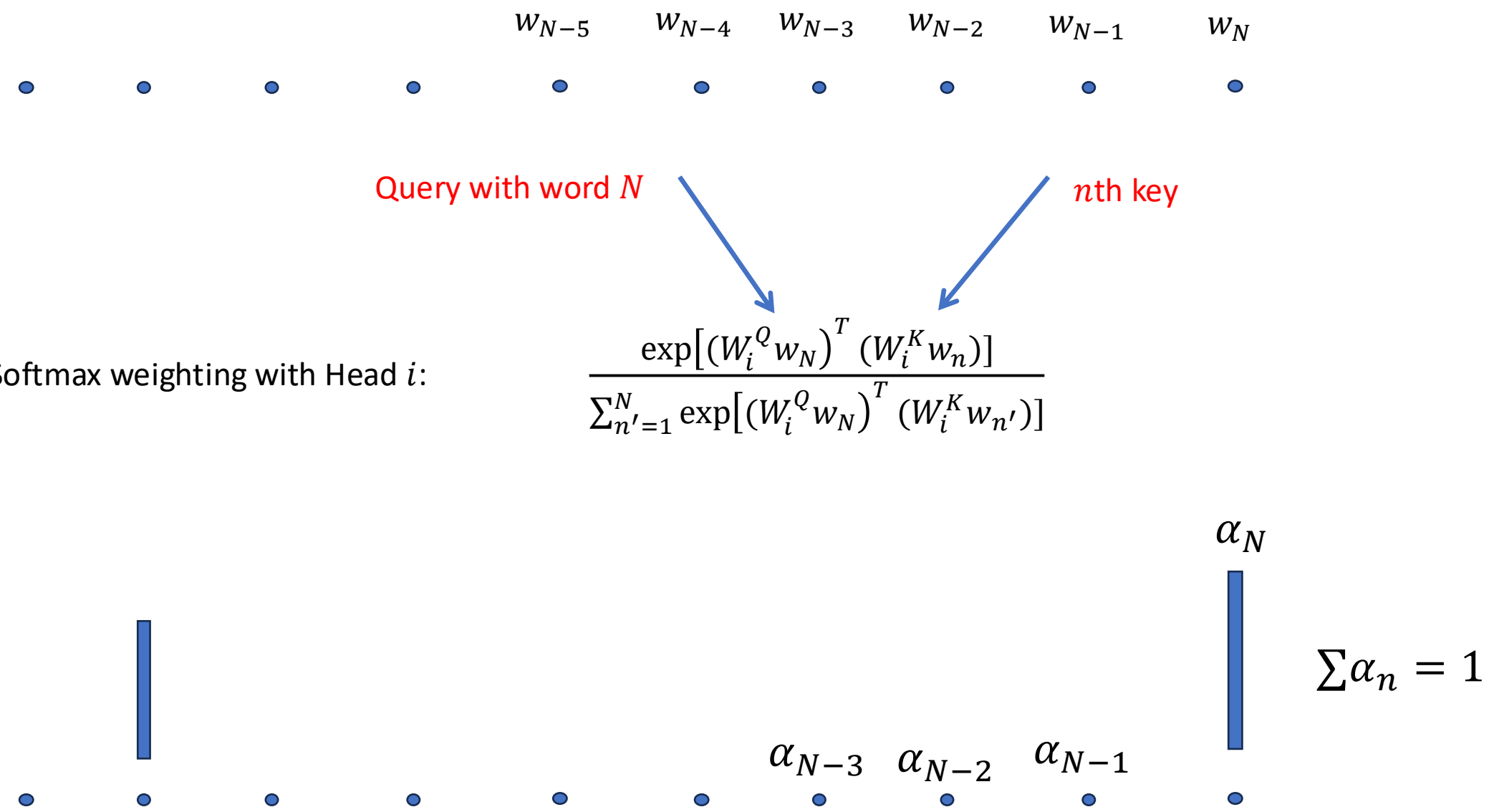
$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

i-th attention head

- W_i^Q and W_i^K designed (learned) such that $W_i^Q p_n \approx 0_d$ and $W_i^K p_n \approx 0_d$ and therefore position <u>not</u> emphasized for query-key match, but rather meaning relatedness emphasized

- W_i^Q and W_i^K learned such that rows of these matrices emphasize components of e_n (word meaning) that are most relevant to relatedness, recalling that components of e_n contain meaning

- W_i^V learned such that $W_i^V p_n \approx 0_d$ and $W_i^V e_n \neq 0_d$ and rows of W_i^V emphasize particular aspects of word meaning, of "most relevance", e.g., query may emphasize if word a noun, and key emphasizes if word a verb, or adjectives related to nouns



Softmax weighting with Head i:

$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

$$\sum \alpha_n = 1$$

w_{N-5} w_{N-4} w_{N-3} w_{N-2} w_{N-1} w_N

Query with word N

nth key

Softmax weighting with Head i:

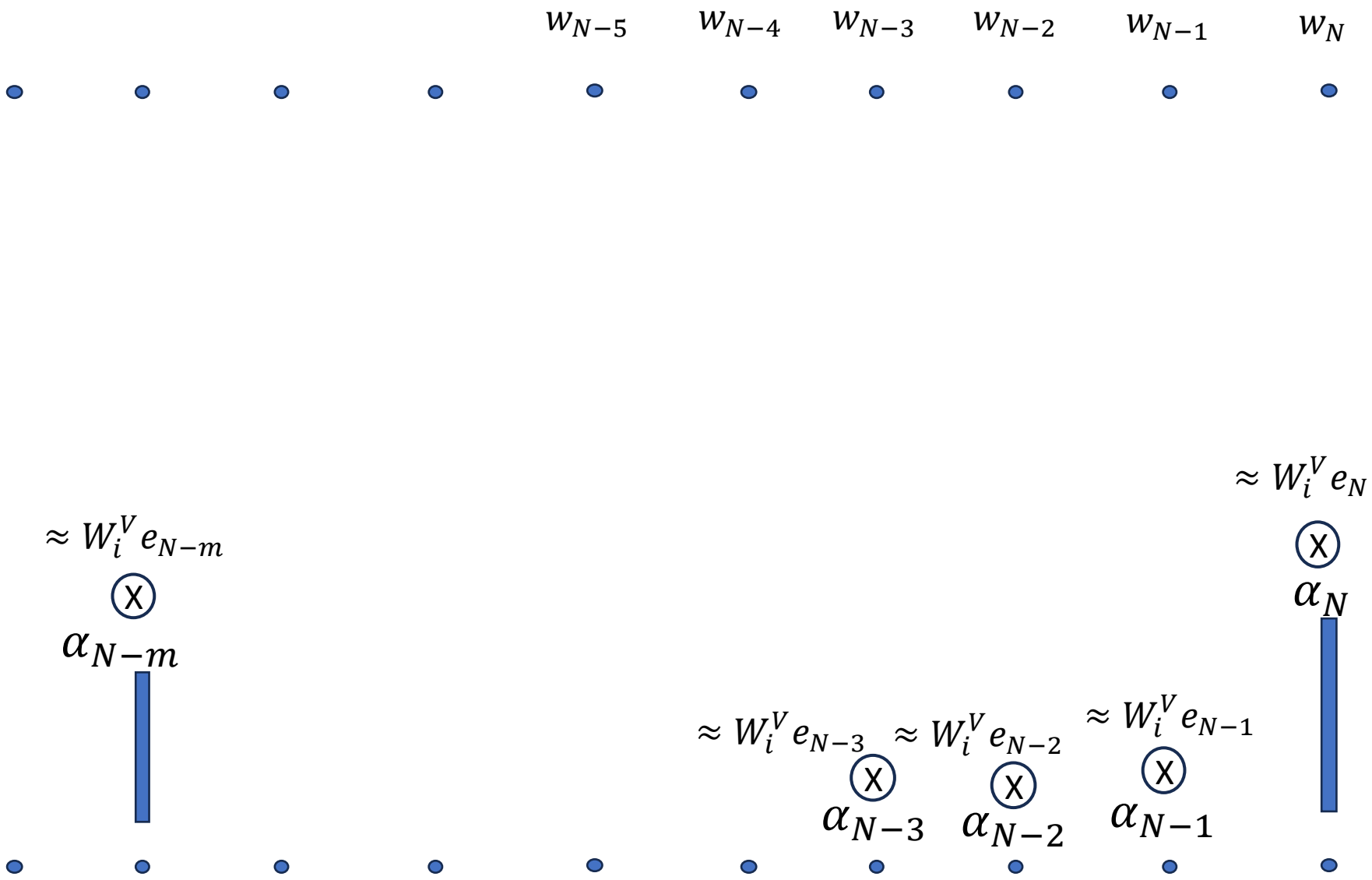
$$\frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

α_N

Query-key match by *meaning*, <u>not</u> position

$\sum \alpha_n = 1$

α_{N-3} α_{N-2} α_{N-1}


w_{N-5}
w_{N-4}
w_{N-3}
w_{N-2}
w_{N-1}
w_N
$\approx W_i^V e_{N-m}$
α_{N-m}
$\approx W_i^V e_{N-3}$
α_{N-3}
$\approx W_i^V e_{N-2}$
α_{N-2}
$\approx W_i^V e_{N-1}$
α_{N-1}
$\approx W_i^V e_N$
α_N

$$\sum \alpha_n = 1$$

w_{N-5} w_{N-4} w_{N-3} w_{N-2} w_{N-1} w_N

$$\alpha_N = \frac{\exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_n\right)\right]}{\sum_{n'=1}^{N} \exp\left[\left(W_i^Q w_N\right)^T \left(W_i^K w_{n'}\right)\right]}$$

Emphasizes <u>meaning</u> of possibly distant words

No impact on positional embedding of w_N

With skip connection: $w_N \rightarrow w_N + \sum_{k=0}^{N} \alpha_{N-k} W_i^V e_{N-k}$

$\approx W_i^V e_{N-m}$

$\otimes$

α_{N-m}

$\approx W_i^V e_{N-3}$ $\approx W_i^V e_{N-2}$ $\approx W_i^V e_{N-1}$ $\approx W_i^V e_N$

$\otimes$ $\otimes$ $\otimes$ $\otimes$

α_{N-3} α_{N-2} α_{N-1} α_N

$$\sum \alpha_n = 1$$

More on Multi-Head Attention & Positional Embeddings

- Recall the word embedding plus positional embedding, for word n in the sequence:

$$w_n = e_n + p_n$$

- The multi-head attention acts on this sum as (because of linearity), for head i:

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

More on Multi-Head Attention & Positional Embeddings

$$w_n = e_n + p_n$$

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- The value heads, W_i^V almost always emphasize the meaning of words, ignoring position
- Different W_i^V may emphasize different aspects of word meaning

More on Multi-Head Attention & Positional Embeddings

$$w_n = e_n + p_n$$

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- The query and key heads, W_i^Q and W_i^K may emphasize meaning or position
- Different heads may emphasize a different mix of *both* position and meaning

More on Multi-Head Attention & Positional Embeddings

$$w_n = e_n + p_n$$

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- Different attention heads learned for each layer of the deep network, allowing different kinds of emphasis
- These attention heads constitute a large fraction of the parameters that are learned

More on Multi-Head Attention & Positional Embeddings

$$w_n = e_n + p_n$$

$$W_i^Q w_n = W_i^Q e_n + W_i^Q p_n$$

$$W_i^K w_n = W_i^K e_n + W_i^K p_n$$

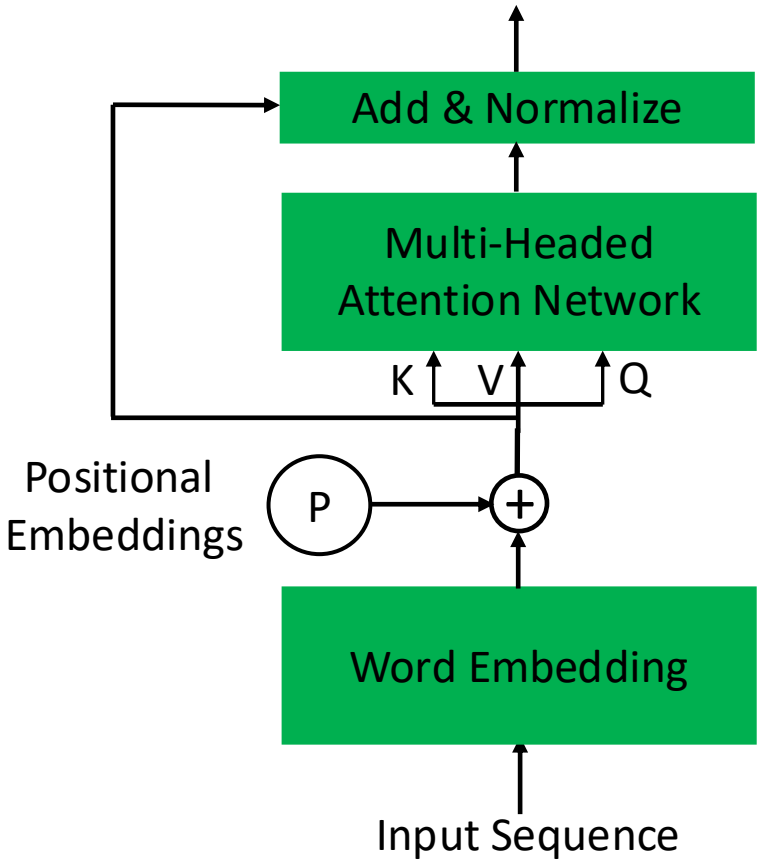
$$W_i^V w_n = W_i^V e_n + W_i^V p_n$$

- Recall that output of all heads concatenated, and combined via a linear element

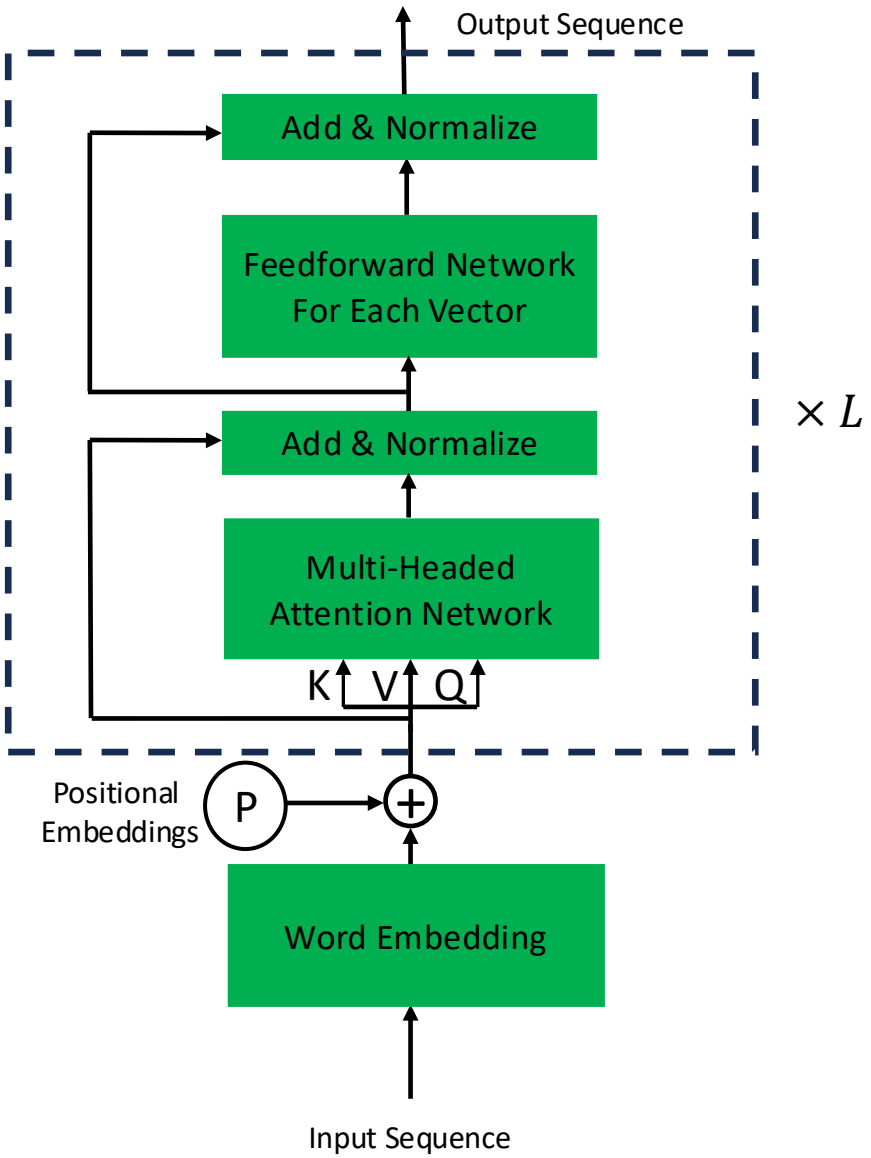
$$\text{output}_i = \text{Attention}(W_i^K K, W_i^V V, W_i^Q Q)$$

$$\text{Multi} - \text{Headed Output}_i = \text{Concat}(\text{output}_1, \dots, \text{output}_h) W^O$$

One Attention Layer

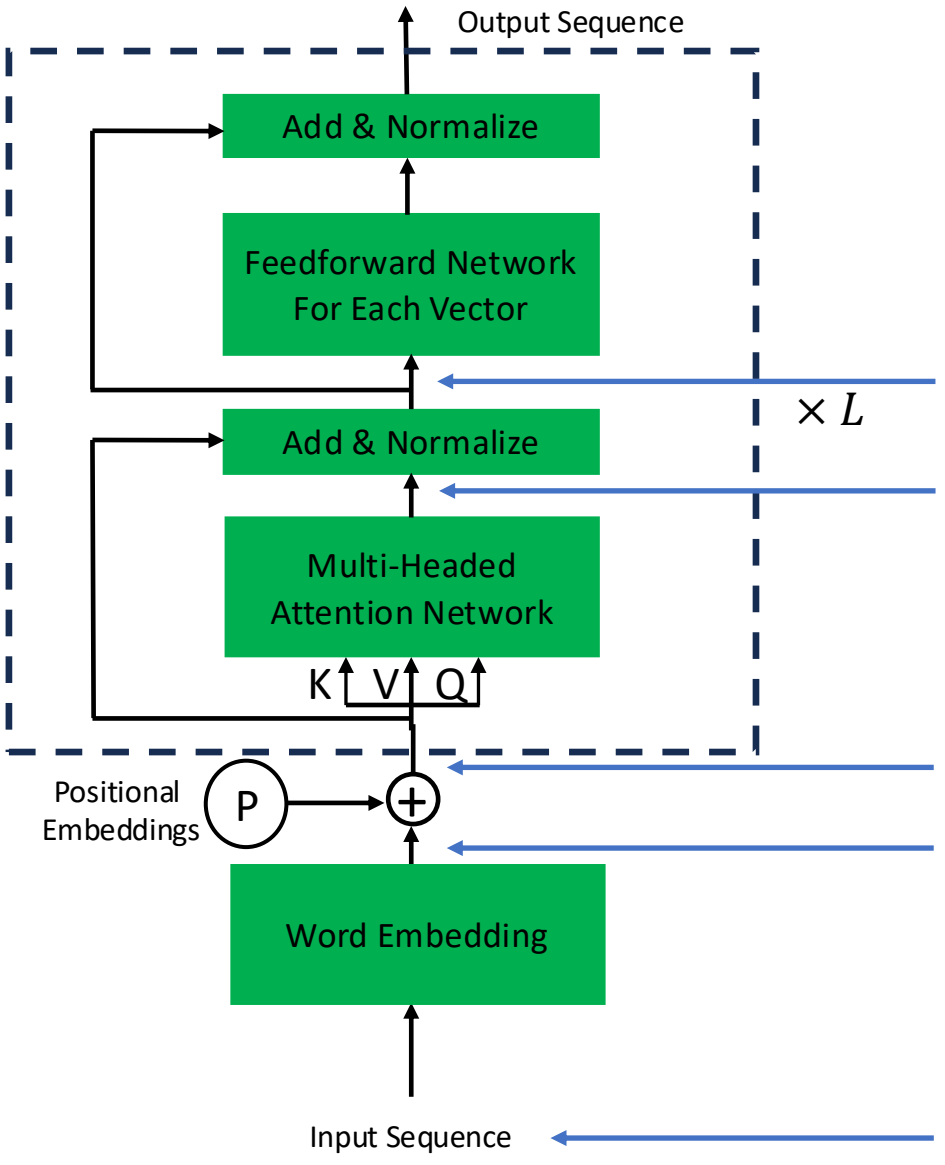


Modification 3: Feedforward Neural Network Transformation



- There are in general different FF parameters at each layer
- In general, different attention heads at each layer
- At a given layer, the same FF network acts independently and identically on each of the vectors output from multi-headed attention

Sequence Encoder



$\{n(n(c_1 + \tilde{c}_1) + f(n(c_1 + \tilde{c}_1))), \dots, n(n(c_N + \tilde{c}_N) + f(n(c_N + \tilde{c}_N)))\}$

$\{f(n(c_1 + \tilde{c}_1)), \dots, f(n(c_N + \tilde{c}_N))\}$

$n(\cdot)$: normalize vector

$\{n(c_1 + \tilde{c}_1), \dots, n(c_N + \tilde{c}_N)\}$

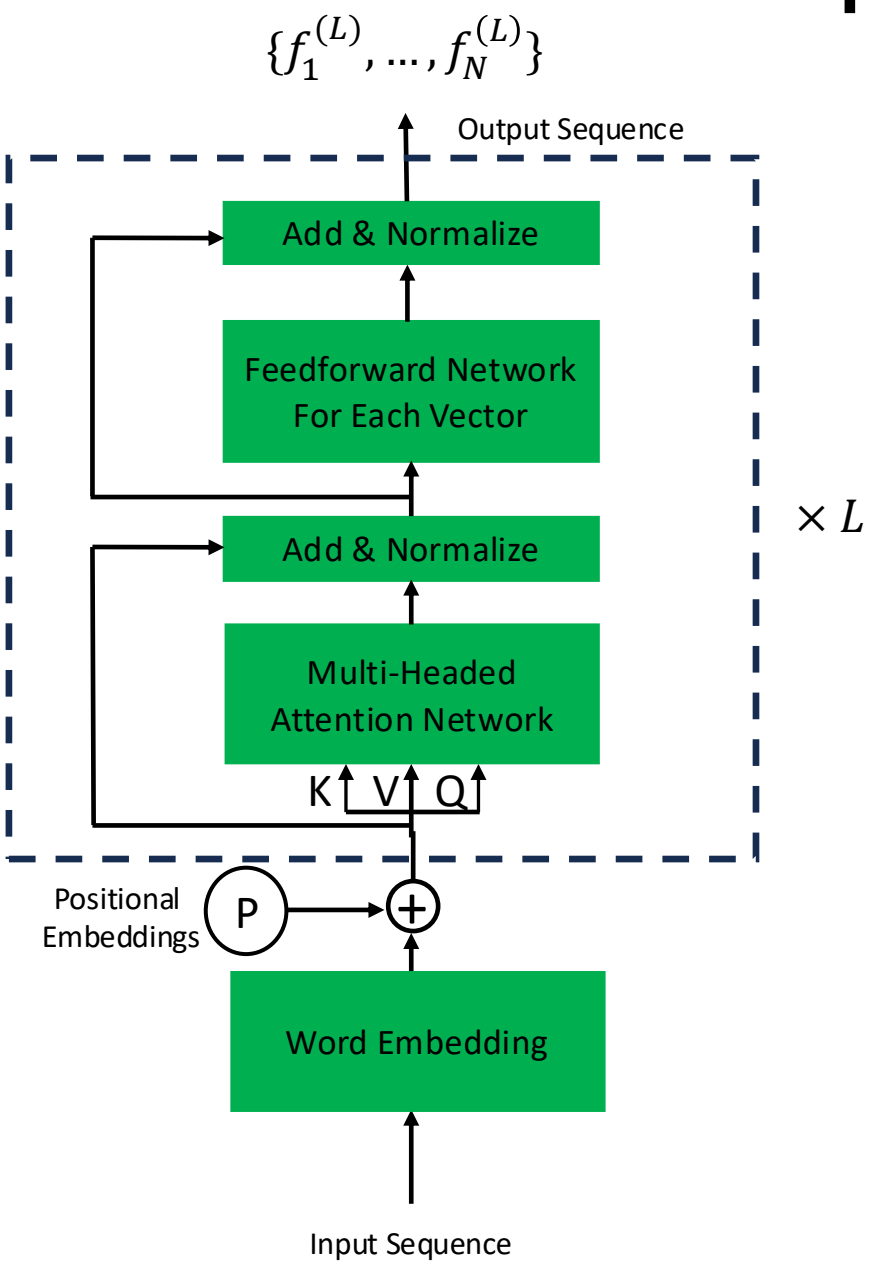
$\{\tilde{c}_1, \dots, \tilde{c}_N\}$

$\{c_1 + p_1, \dots, c_N + p_N\}$

$\{c_1, \dots, c_N\}$

$\{w_1, \dots, w_N\}$

Text Generation



- The words for the prompt text are the first input sequence
- Assume that the prompt has N tokens
- At the last layer of the network, let

$$\{f_1^{(L)}, \dots, f_N^{(L)}\}$$

  represent the vectors output for token positions 1 through N, where N-th word is the last in the sequence
- Goal is to predict the next word, word $N + 1$

Text Generation

- Recall that we have a set of learned embedding vectors $(w_1, \dots, w_V)$ for the V tokens in our vocabulary with each $w_v \in R^d$

- Assume that $f_N^{(L)} \in R^d$

- Additional parameters manifested by matrix $M \in R^{d \times d}$ acting after the last layer

- Define the vector output $u = M f_N^{(L)}$

Softmax Probability of Next Word

- Define the vector output $u = M f_N^{(L)}$

- The probability of token $v \in \{1, \dots, V\}$ being the next word is modeled as

$$p(v|\text{prompt}) = \frac{\exp[w_v^T u]}{\sum_{v'=1}^{V} \exp[w_{v'}^T u]}$$

Training via Cross-Entropy Loss

- Let all Transformer parameters be represented as θ, including the token embedding vectors

- The probability of token $v \in \{1, \dots, V\}$ is

$$p_\theta(v|\text{prompt}) = \frac{\exp[w_v^T u]}{\sum_{v'=1}^{V} \exp[w_{v'}^T u]}$$

- Assume we have a collection of text prompts, $\left\{\text{prompt}^{(j)}\right\}_{j=1,J}$ and let $w_{N+1}^{(j)}$ represent the true next word connected to prompt j

- Learn θ by seeking to minimize the cross-entropy loss for prediction of the next word

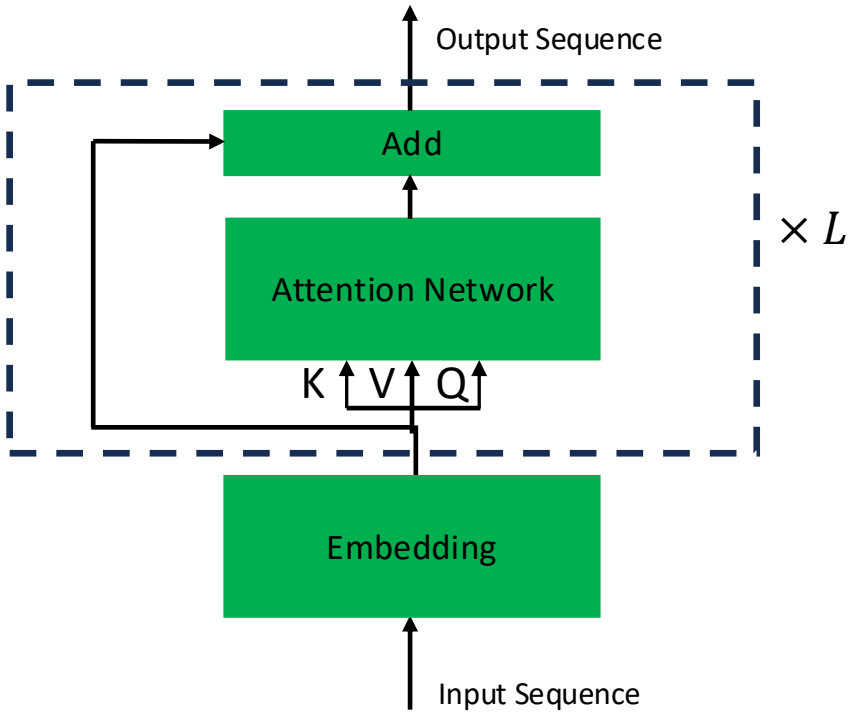
Training via Cross-Entropy Loss

- The probability of token $v \in \{1, \dots, V\}$ is

$$p_\theta(v|\text{prompt}) = \frac{\exp[w_v^T u]}{\sum_{v'=1}^{V} \exp[w_{v'}^T u]}$$

- Assume we have a collection of text prompts, $\{\text{prompt}^{(j)}\}_{j=1,J}$ and let $w_{N+1}^{(j)}$ represent the true next word connected to prompt j

$$\hat{\theta} = \text{argmin}_\theta \; -\sum_{j=1}^{J} \log \, p_\theta(v = w_{N+1}^{(j)}|\text{prompt}^{(j)})$$

- Design the Transformer parameters such that the model has a high likelihood of predicting the true next word, across all prompts in the corpus

Sequence Encoder